

Shaft Sinking Preparation

15/05/2020

Design I

- *Shaft is well like structure, excavated in earth through soil and rock to make entry to ore deposit or other facilities.
- *Shaft can be vertical or incline.
- *Shaft can start from surface or in underground (blind shaft, staple shaft, Sub vertical shaft).
- *Shaft establishes link between mines surface facilities and underground installation.
- *Shaft should have sufficient capacity to take care anticipated future changes / increase in requirement.

Design II

- *Mine shaft, including head frame, winder have to be designed to last life time of mine.
- *Shaft design should reflect adequate flexibility to accommodate different clangs in future.
- *Mine shaft design should consider time for construction to reduce gestation time for starting production.

Design III

- *Shaft layout / design should be such that “Domino” effect is eliminated in case of breakdown / failure.
- *Shaft plant design should eliminate wastage, bottle neck in system and perform with efficacy in changed environment.
- *All related information, data are to be gathered as input before designing shaft plant.
- *If possible, in formations and data are to be verified from different sources.

Design IV

- *Different alternative design, solutions are to be developed and examined before selecting best suitable design.
- *It is better to over design shaft than de-bottlenecking or sinking a new shaft to handle production in future.
- *Geological information like Rock strength detail, geotechnical parameters are to be collected in advance.
- *Ground water level together with water head calculation and inflow of water.

Design V

- *Danger of chemical contamination, pH etc.
- *Shaft diameter to be decided based on duty of shaft.
- *Shaft sinking technology to be fixed.
- *Detail of shaft lining, thickness, material etc. has to be determine well in advance.
- *Shaft collar depth, foundation, thickness, construction material, number of opening with their functions, dimensions and elevation.
- *Shaft sump, its depth, structure, construction, pumping arrangement, dimensions, elevations.
- *Survey data for shaft and other establishment.

Design VI

- *Ground and water pressure acting on lining resulting thickness of shaft wall inset with dimension.
- *Air flow capacity of shaft has to be considered based on heoghest air requirement of mine.

Design V

- *Time for construction to be calculated, use of PERT, CPM, MS Project etc till production resumes through shaft.
- *Cost specification has to be detailed.
- *Detail engineering and construction, fabrication drawing for each equipment, part etc to be made ready.
- *Construction, fabrication procedure are to be prepared well in advance and communicated down the line.

Shaft Design A

- *Cross section of shaft (area) is decided based on duty of shaft to perform.
- *Number of man power to move through shaft.
- *Volume of material to be hoisted through shaft.
- *Type of fittings (rope guide, rail guide, cage hoisting, skip hoisting etc) in shaft and mode of hoisting.
- *Largest size material requirement to be handled in shaft.
- *Requirement of compressed air, pump delivery line, ladder ways, power cables etc.

Shaft Design B

- *Different shape of cross section are practiced.
- *Circular
- *Squire
- *Rectangular
- *Elliptical
- *Oval
- *Rectangular with elliptical sides.

Shaft Design C

- *Single multipurpose shaft is becoming more common, especially for deep mines.
- *Ventilation shaft is normally located at peripheral area.
- *Cost for shaft sinking for large diameter shaft does not go up proportionately.
- *When mine ore body is elongated and spread over on large distance, additional shaft can be economical.

Shaft Location Surface I

- *Shaft location is decided considering following features at surface.
- *Surface Building and structure.
- *Ancillary work shop at surface.
- *Pipe line for different facilities.
- *Equipments, viz compressor, substation etc.
- *Electric Power Cables
- *Size of stock pile area, mine supply yard capacity and location.

Shaft Location Surface II

- *Ponds.
- *Roads.
- *Rail.
- *Other required supply line Rope Ways.
- *Equipments, viz compressor, substation etc.
- *Electric Power Cables.
- *Ore Process Plant.
- *Surface Topography.
- *

Shaft Location Surface III

- *Proximity to service like power, water, compressed air line.
- *Location of rail and road communication.
- *Proximity of population.
- *Regulatory and environment constraints.
- *Ease of access to personnel.
- *Drain and drainage facilities.
- *Place for ore & Waste dumping / stacking yard.
- *

Shaft Location Surface IV

- *Water runoff ditches.
- *Shaft caller should be 3 to 5 meter above highest flood level is desirable.
- *Space for tailing / sand stowing facilities.
- *Space for Head Frame, Winders house, vehicle garage etc.
- *Space for substations, garage, storage for cage, skip etc.

Shaft Location Surface V

- *Pit Top Bath (Bath House).
- *Topography of area, steep hill slope, surrounded by river etc.
- *

Shaft Location Under Ground I

- *Mineral.
- *Ore Body configuration.
- *Geological structure.
- *Geo mechanical properties of ore, HW & FW etc.
- *Underground workshop for repairing.

Shaft Location Underground I

- *Shaft can be located on FW, HW of ore body or can pass through ore body in between.
- *Centre of gravity (CG) of ore body is considered for locating shaft to economies on underground transport.
- *Shaft located in FW or ore body is least likely to get disturbed by mining activities.
- *As per as possible, locating shaft at HW should be avoided to minimize disturbance by mining activities in future.

Shaft Inset I

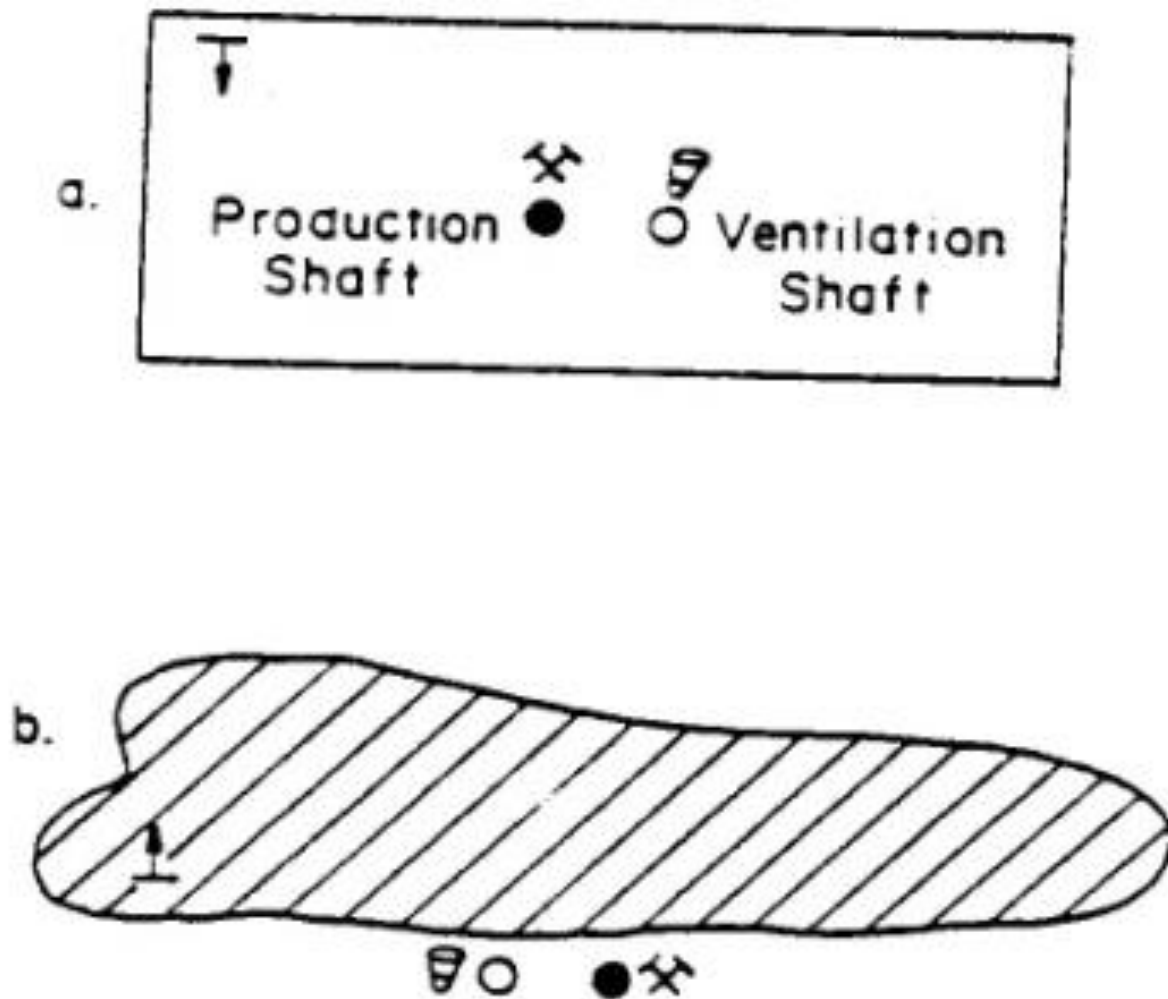
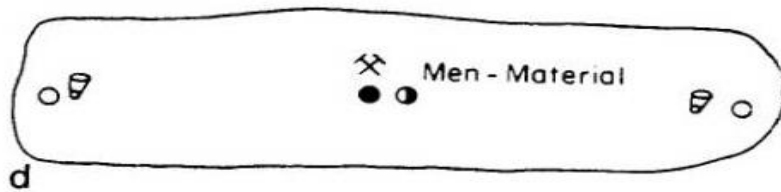
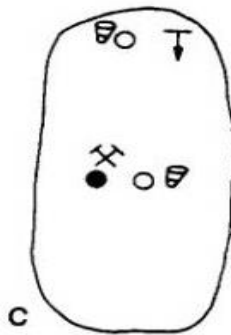
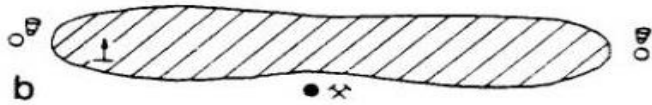
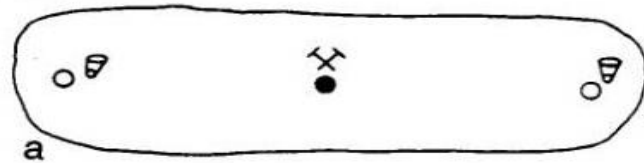


Fig. 17.4.2. Shaft locations for a two-shaft mine (a) centrally located and (b) on the footwall side of a steeply dipping deposit with its major axis in the strike direction. (After Jawien and Suchan, 1980.)

Shaft Inset I

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Fig. 17.4.4. Shaft locations for three- and four-shaft mines. Three-shaft layouts: (a) elongated deposit with major axis parallel to strike; (b) steeply dipping deposit with major axis parallel to strike; (c) elongated deposit with major axis normal to strike. Four-shaft layouts: (d) elongated deposit with major axis parallel to strike (after Jawien and Suchan, 1980).

Shaft Components I

- *Mine shaft includes following structure and fabrication.
- *Shaft collar.
- *Ventilation air ways.
- *Pumping & mine water line.
- *Water collection structure.
- *Communication & power line.
- *Shaft transport system.

Shaft Components I

- *Ore Bin (Surge Bin) to stock ore at bottom of shaft for loading ore in to skip.
- *Ore bin lowers potential disruption and to maintain a high continuity of mining operation, should a problem develops in shaft hoisting system.
- *Electric power cable is placed in shaft.
- *Consider possibility of taking down power cable through bore hole.
- *Power cable in bore hole eliminates possibility of cable damage moving object or falling material in shaft.

Geology & Geomechanics I

- *Reliable geological information to be collected by core drilling near shaft site.
- *Bore hole should not be drilled in side proposed shaft or effected area.
- *Exploration hole should be drilled 10 to 30m from shaft centre (axis).
- *This distance ensures that result from drill hole corresponds to condition of shaft.
- *In case, hole is drilled with in shaft circumference, it would may have detrimental effect on sinking operation.

Geology & Geomechanics II

- *Hole may interact water bearing levels, resulting an erroneously high hydrostatic pressure.
- *These geological and geomechanical information are to be used for shaft design.
- *Geological and geomechanical information's are collected from core of drill hole.
- *This information's are to be collected throughout depth of shaft and of all different formations the shaft is passing through.

Geology & Geomechanics II

- *Geomechanical information's should include following information along with other information.
- *Description of geology, Granulometric analysis of unconsolidated strata, angle of repose, internal friction, specific gravity, porosity, fracture description, Rock Quality Designation (RQD), inclination of strata.
- *Geophysical logging of neutron density, self potential resistivity.

Geology & Geomechanics III

*Some portion of core and water sample should be preserved in their natural site for testing in future, if necessary for future comparison.

Water Flow I

- *Shaft face interacting a water bearing strata will collect water similarly to the way well collect water.
- *Water flow must be controlled by pumping .
- *Pumping creates a depression cone created around shaft.

Water Flow II

*Water flow in to shaft can be calculated.

q >> inflow in m^3/day into shaft through walls (free water flow).

K >> coefficient of filtration in m/day .

H >> head of water table in m .

h >> head of distress water table in m .

R >> radius at depression cone in m

r >> radius of shaft in m

s >> $H-h = m$, deprecation in m

1. An inflow q in m^3/day into the shaft through the walls (free water level) is calculated from

$$q = 1.36 K \frac{H^2 - h^2}{\log (R/r)} \quad (17.4.1)$$

2. An inflow into the shaft through the walls for water level with a head is

$$q = 2.73 K m \frac{H' - h'}{\log (R/r)} \quad (17.4.2)$$

here m is thickness of the nonpermeable strata.

3. Inflow through a flat bottom is

$$q = 4K r s \quad (17.4.3)$$

4. Inflow into the shaft through the shaft walls in fissured rocks and free water level (simplified formulas) is

$$q = 2\pi K H \sqrt{r s} \quad (17.4.4)$$

5. Inflow for this same condition but with a head is

$$q = 2\pi K m \sqrt{r s} \quad (17.4.5)$$

Drawdown Ground Water I

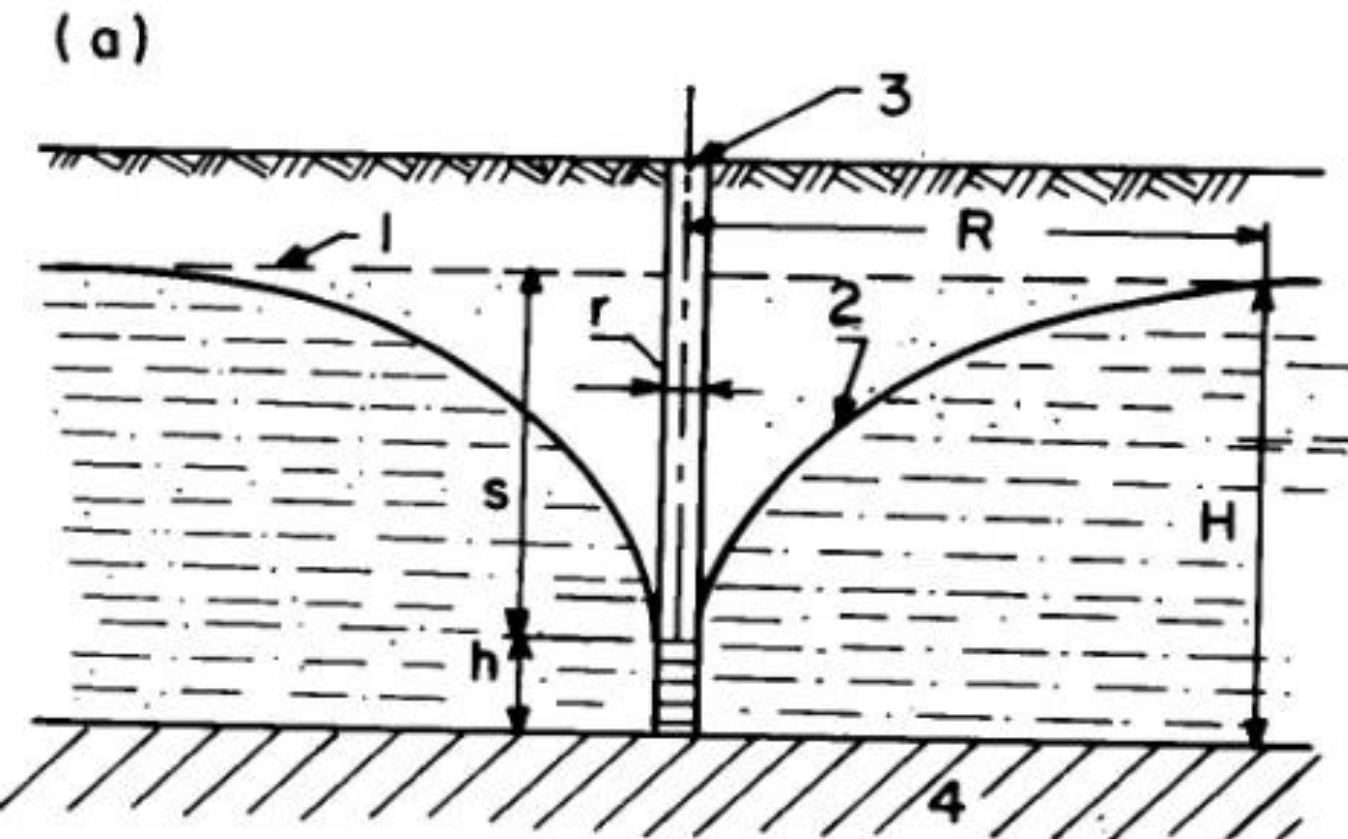


Fig. 17.4.1. Schematic of drawdown

- a) water-bearing strata with free ground level
1. initial groundwater level
 2. drawdown water level
 3. depression well
 4. nonpermeable strata

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- b) water-bearing strata with head
 1. nonpermeable upper strata
 2. water-bearing strata
 3. initial head of groundwater
 4. drawdown water level

Water Flow I

*Water flow to mine shaft through wall

Shaft Diameter I

- *Lateral cross section of shaft is determine graphically.
- *All fittings of shaft is placed in plan with required clearance.
- *Shaft cross section shall allow required air in underground with permitted velocity.

Shaft Lining I

- *Shaft lining support wall of excavation.
- *Shaft lining support shaft fittings.
- *Shaft lining is minimum 200 mm but 300 mm is common thickness.
- *This thickness is required to grout different fitting in shaft wall.
- *PCC is most commonly used shaft lining material.
- *PCC offer smooth surface reducing air resistance, make sinking faster and permits taking full advantage of structural feature of concrete.

Shaft Lining II

- *PCC lining is cheaper.
- *Other materials of shaft lining are brick, concrete block, steel tubing etc.
- *PCC placement is mechanized, resulting high sinking rate.
- *Strength of lining can be adjusted (20 to 50 Mpa).
- *Water tight lining can be achieved for aquifer with moderate head.
- *In difficult ground conditions, concrete mantel and welded steel lining are applied.

Shaft Lining III

- *In case of sinking through shaft pillar, then a special elastic shaft lining is designed, consisting two layers separated most often by bitumen envelop.
- *This construction prevents deformation of inner structure lining caused by shifting of subsidizing rock strata.

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Shaft Collar I

- *Shaft collar is upper part of shaft extending to first footing.
- *Shaft collar must be anchored in competent rock.
- *Diameter of shaft collar is same as that of shaft.
- *Depth, cross section, thickness of shaft collar depends up on character of over burden, rock hydrologic conditions, water and ground presser, sinking method, load of head frame, horizontal reaction from foundations of nearby structure.

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Shaft Collar II

- *Shaft collar for production shaft should have provision of accommodating water pipes, compressed air pipe line, electrical and communication cable, ladder or any other fitting in shaft.
- *Shaft collar for ventilation intake shaft to be designed to have provision of heated air intake in winter or cooled air in summer, if required.
- *Ventilation shaft collar should have ventilation drift conducting exhaust or intake air to main fan if located on surface.

Shaft Collar III

- *Shaft collar should have additional outlet / fittings to meet special need of shaft.
- *At hoisting shaft , collar must be designed to provide support for head frame.
- *When over burden is of low strength, head frame is to be provided support from shaft collar as well as its own foundation. Composite foundation.
- *Base of head frame can rest on shaft collar well, if provision for this additional load has been made in advance.

Shaft Collar IV

- *Head frame can also be placed on steel beam fastened into collar wall.
- *For installing large winder on top of head frame, additional foundation outside Shaft collar wall may be required.
- *For smaller winder, a round tower in design, similar to extension of shaft, integrated with reinforced concrete collar from certain depth below ground.
- *Thickness of shaft collar to be increased (up to 0.6m, as per design) to serve as foundation of head frame.

Shaft Collar V

- *In present days, use of tower mounted, multiple rope friction winder has been increased.
- *Shaft collar is made of Reinforced concrete.
- *Shaft collar is thick in compared to shaft lining.
- *Shaft collar has to bear additional load.
- *Shaft collar has weakening structural influence of multiple outlet and connection.
- *Shaft collar is exposed to change in temperature, dynamic loading in both vertical and horizontal direction caused by movement of conveyance in shaft.

Shaft Collar VI

- *Beside RCC, other material may be used for shaft collar are concrete block, pre casted reinforced concrete element, high strength break etc.
- *Reinforced concrete is used for shaft collar to take care of large and eccentric dynamic load on collar.
- *Shaft collar usually have a step down shape, with footing foundation at bottom.
- *First step must be deeper than 4 to 5 meter (frost Penetration)

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Shaft Collar VII

- *Exact Shaft Collar depends on ground and loading condition.
- *First collar thickness is 1-2m.
- *Next step is 0.6 – 1m thick (approximately two times the thickness of shaft lining).
- *Wall thickness of third step is in-between ground step and regular shaft lining.
- *Foot of collar is placed in firm strata, 2 to 3m below over burden.
- *Shape of shaft collar is often double conical to better transfer of load to surrounding.

Shaft Collar VIII

- *Influence of foundation of adjacent structure (head frame, silo etc) on shaft collar is evaluated graphically.
- *When distance is less then, additional load must be found from stress distribution diagram.
- *In this situation, collar thickness have to be increased.
- *Shaft collar is to be designed to stand uneven, horizontal and vertical, uneven stress due to eccentric loading of head frame.

Shaft Collar IX

- *While designing shaft collar, it is assumed that there is no friction between lining and surrounding ground.
- *Four head frame legs carry vertical force P_1 , P_2 , & P_4 ,

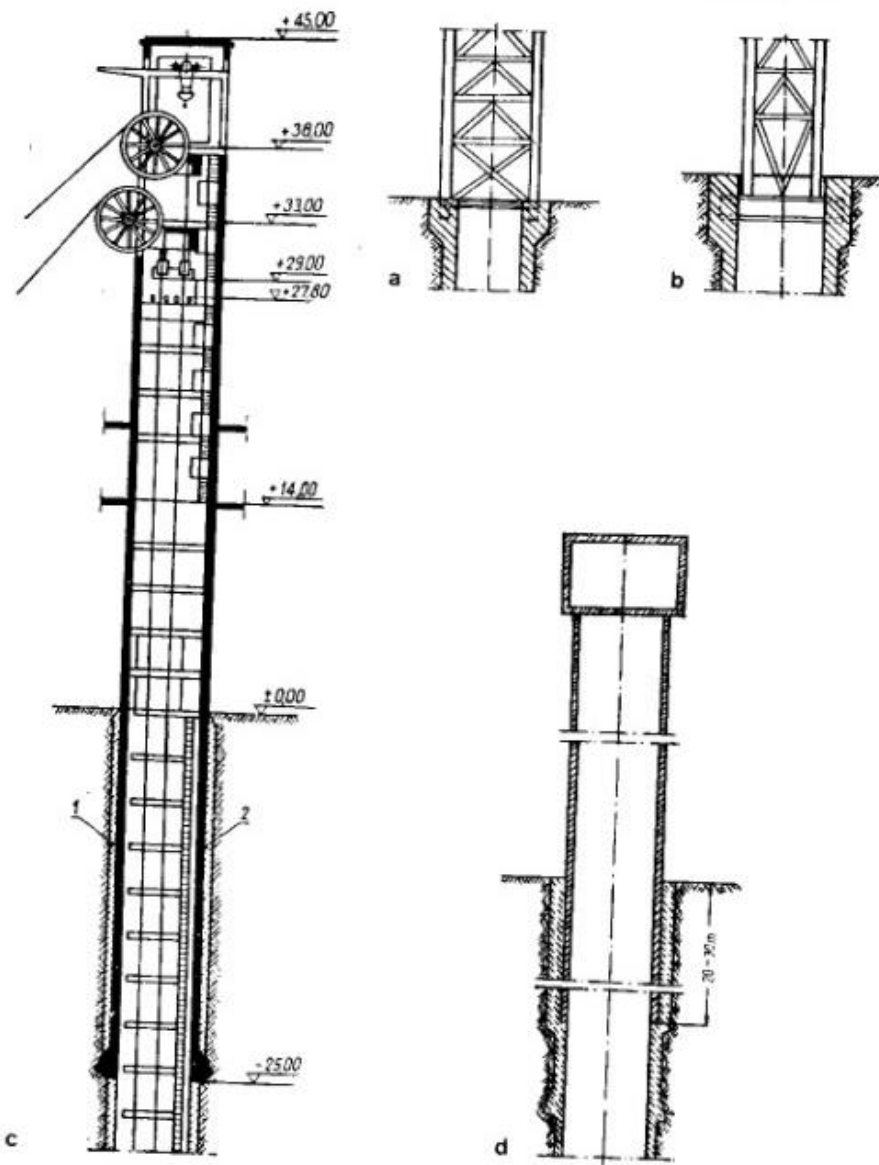


Fig. 17.4.9. Sections through headframe and headgear foundations: (a) foundation on shaft head; (b) foundation on beams located in shaft collar wall; (c) circular headgears founded on shaft head, comprising concrete shaft head and reinforced concrete headgear. Dimensions in meters. (After Unrug, 1985.) Conversion factor: 1 ft = 0.3048 m.

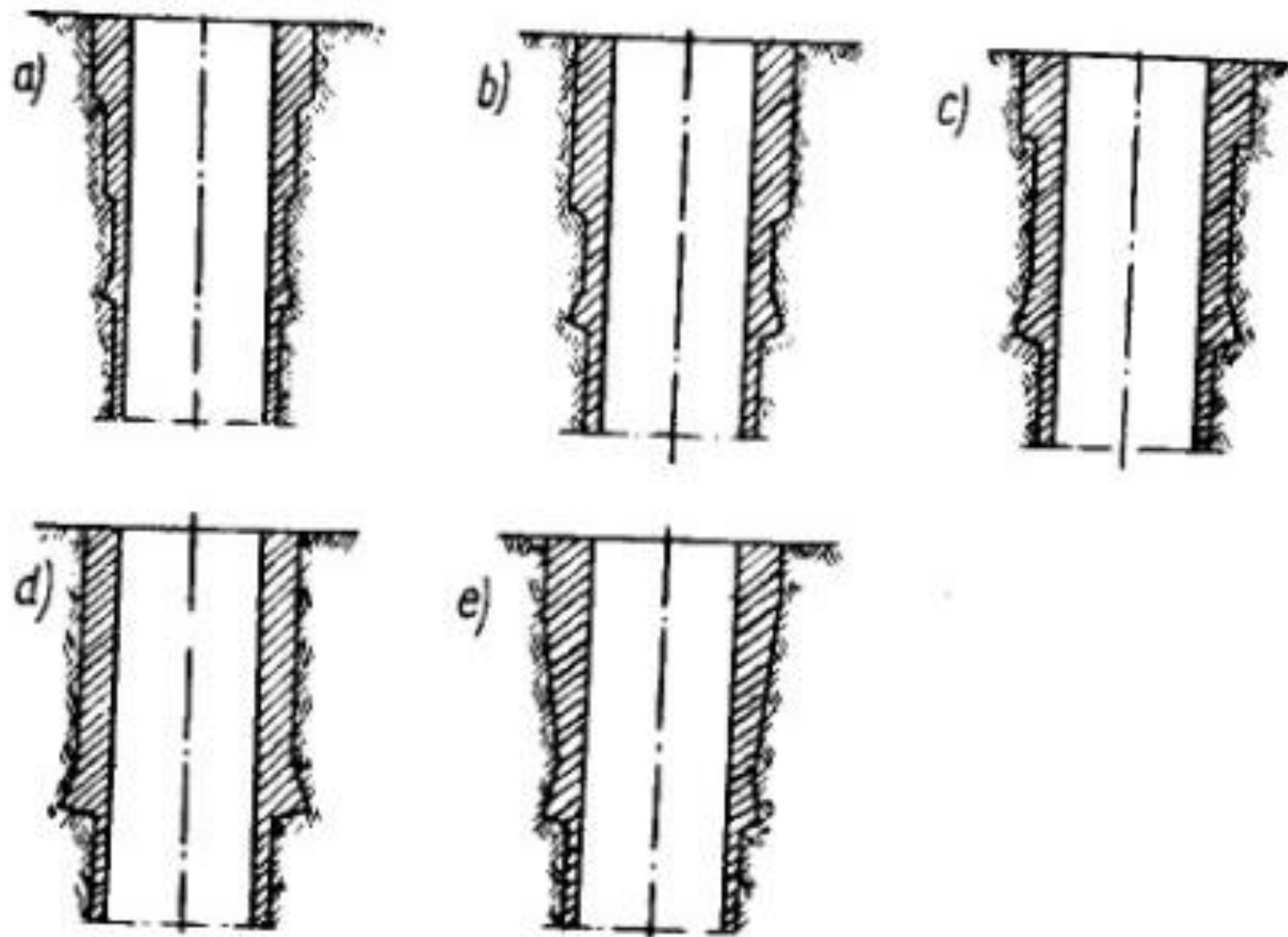


Fig. 17.4.10. Different shapes of shaft collars. (After Unrug, 1985.)

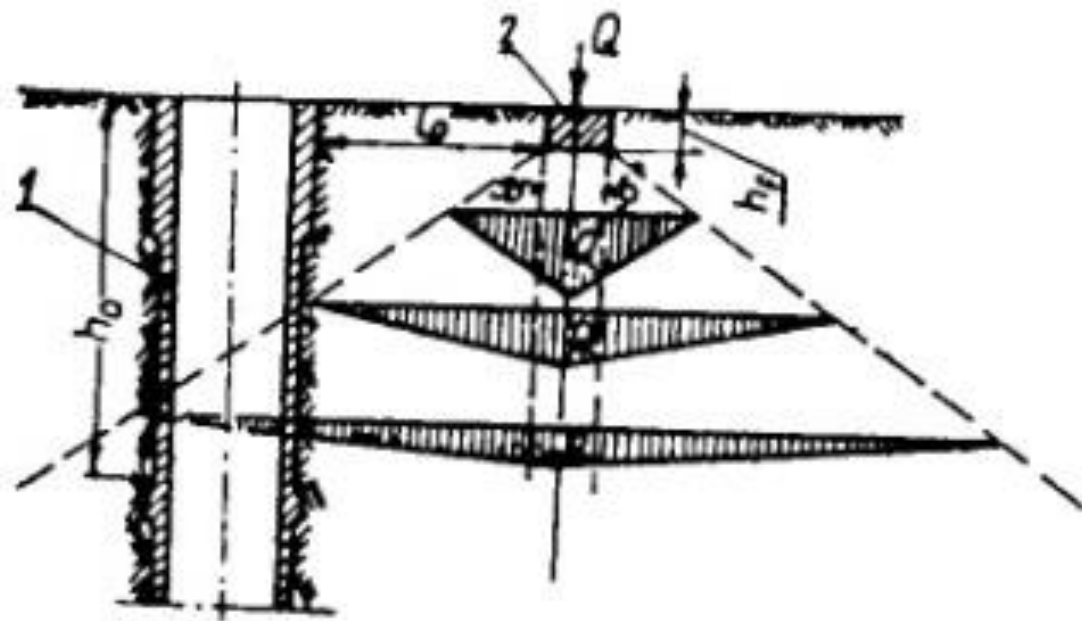


Fig. 17.4.11. Simple geometric method for determining additional load acting on the shaft collar lining from nearby foundations. A zone of influence is defined by lines drawn at 35° from the vertical at the base of the foundation. The effect of additional foundation loading may be considered negligible when the horizontal distance from the shaft to the foundation l_0 is greater than $(h_0 - h_f) \tan 55^\circ$, where h_0 is depth of the shaft collar, and h_f is depth of the foundation. (After Unrug, 1985.)

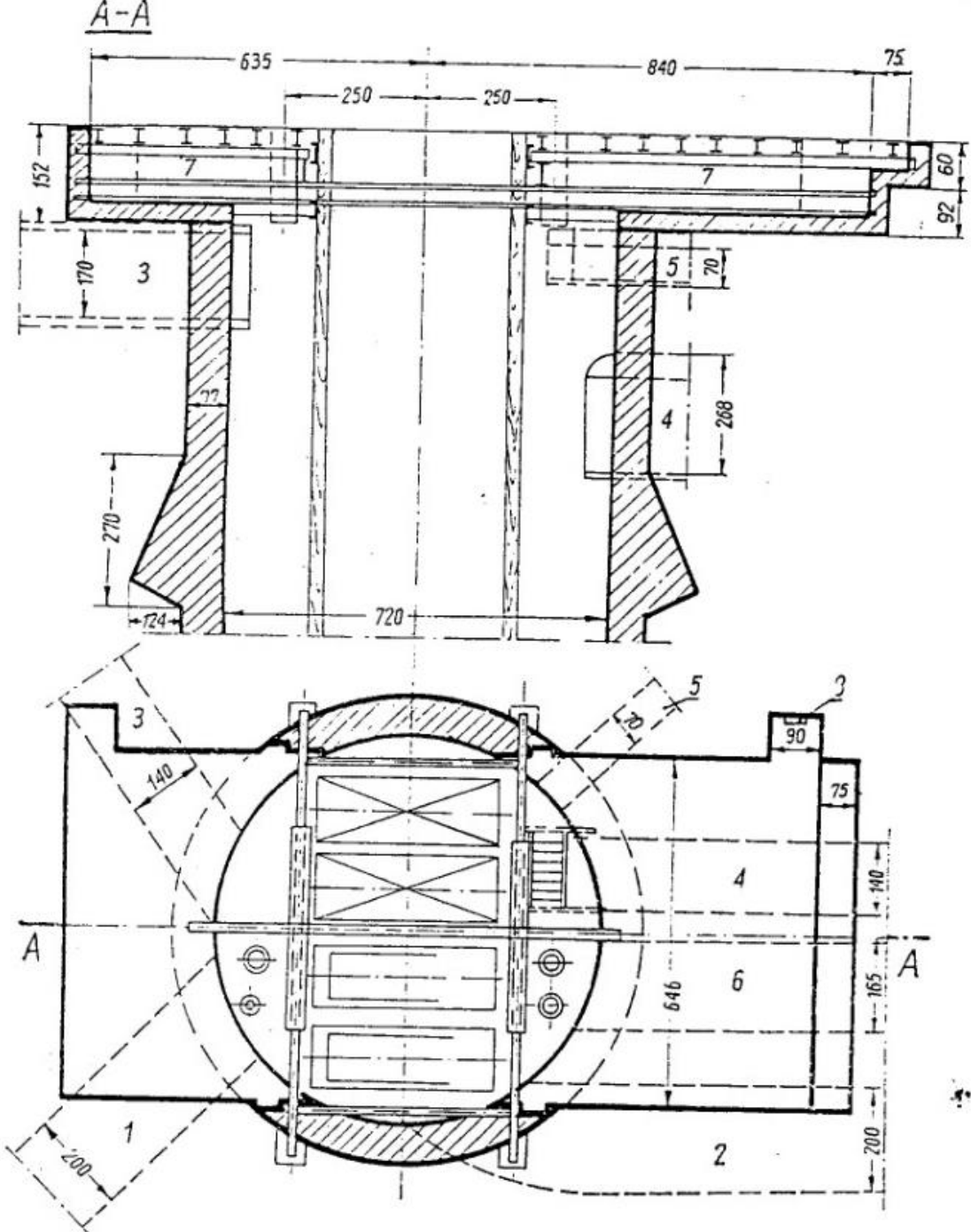


Fig. 17.4.13. Production shaft head 24 ft (7.2 m) in diameter.

- 1. heated air channel I
 - 2. heated air channel II
 - 3. channel for water pipes
 - 4. outlet from ladder compartment
 - 5. cable channel
 - 6. channel for compressed air lines
 - 7. basements for pushers
- Dimensions in cm. Conversion factor:
1 in. = 2.54 cm.

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Shaft Inset I

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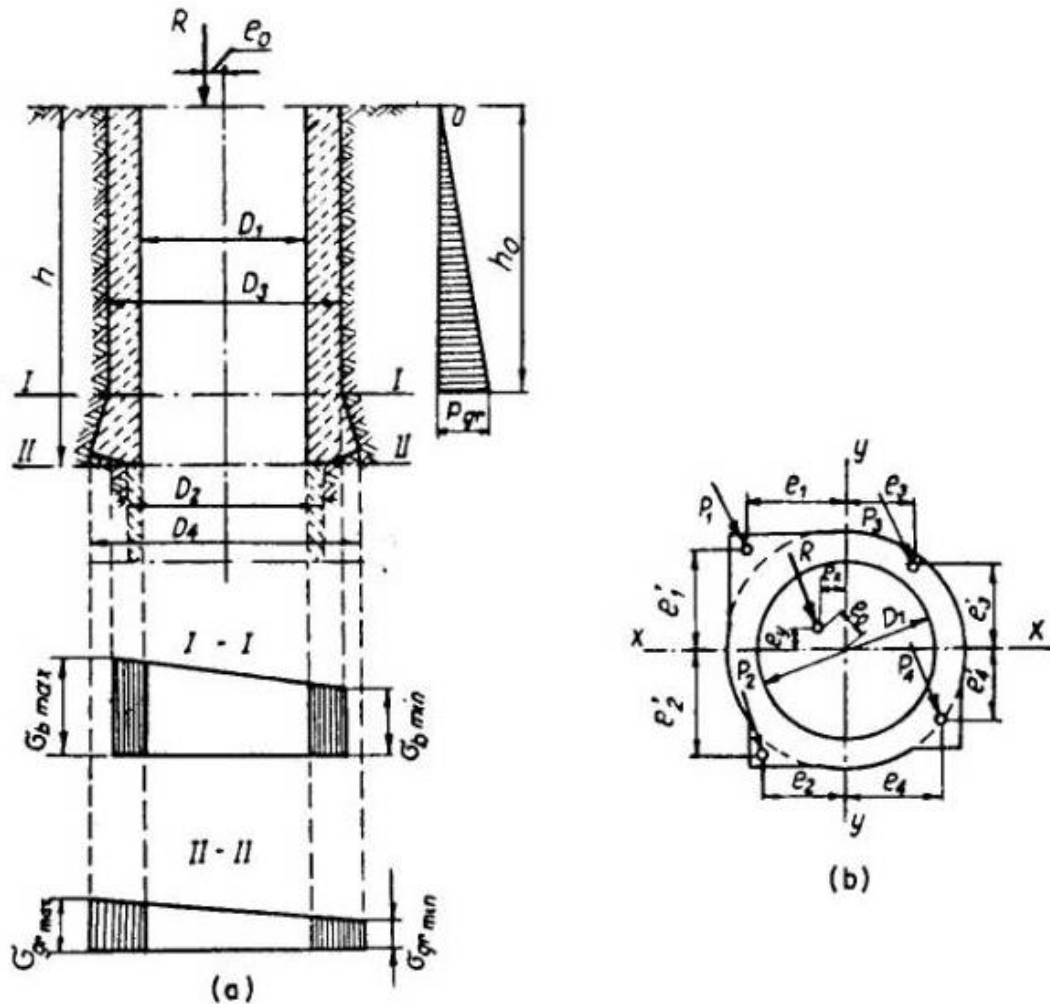


Fig. 17.4.12. Load distribution on shaft collar lining [p —forces, e —coordinates of forces, R —resultant force, D —shaft diameters]. (After Walewski, 1965.)

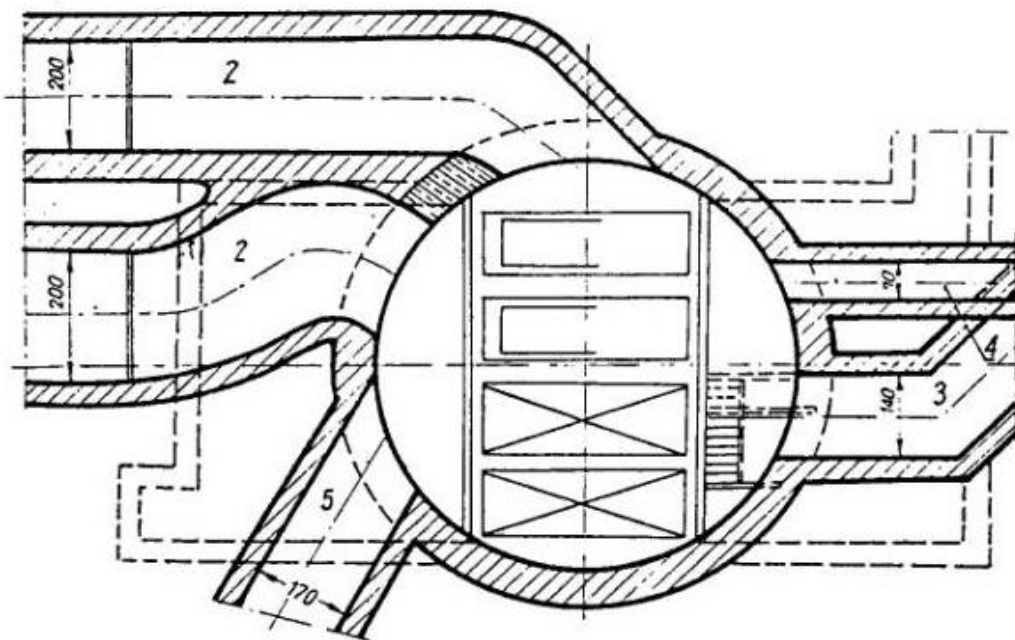
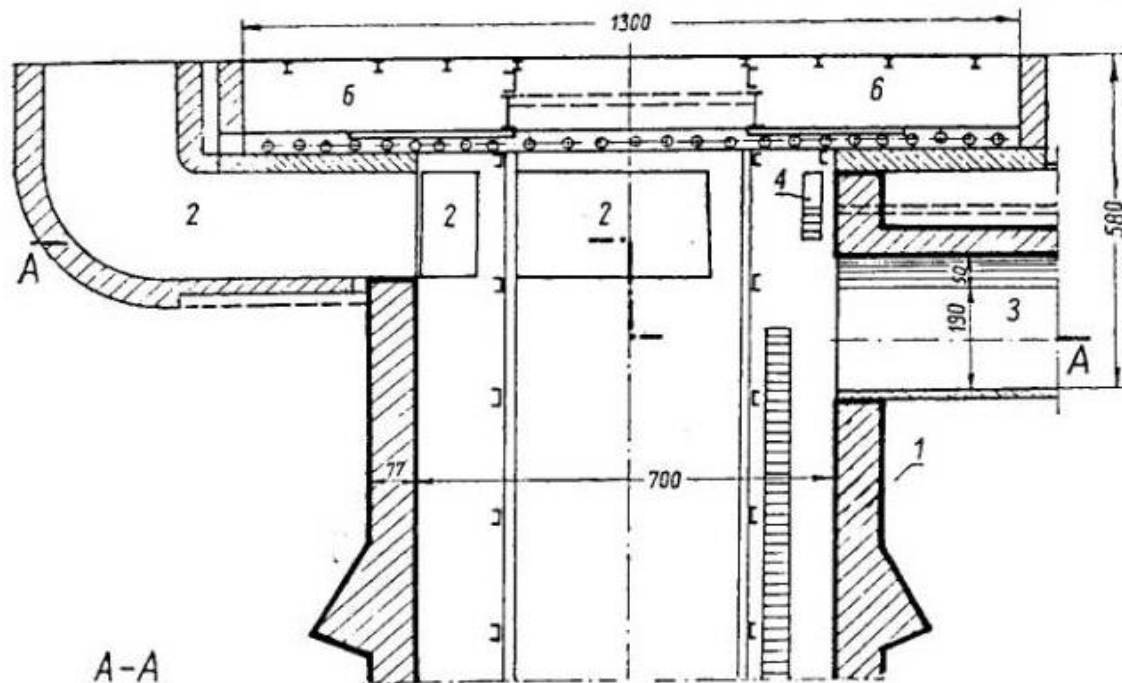


Fig. 17.4.14. Design of production shaft head 23 ft (7 m) in diameter.

1. lining (bricks)
2. heated air channels
3. outlet from ladder compartment
4. cable channel
5. channel for water pipes and compressed air
6. basement for fire closings

Dimensions in cm. Conversion factor:

1 in. = 2.54 cm.

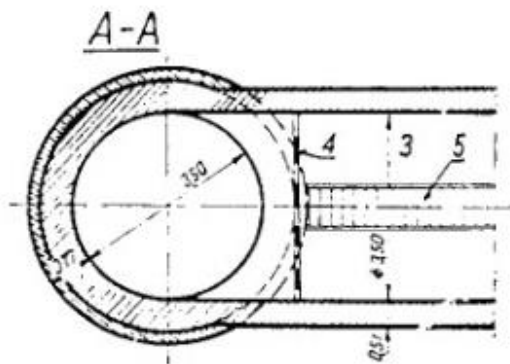
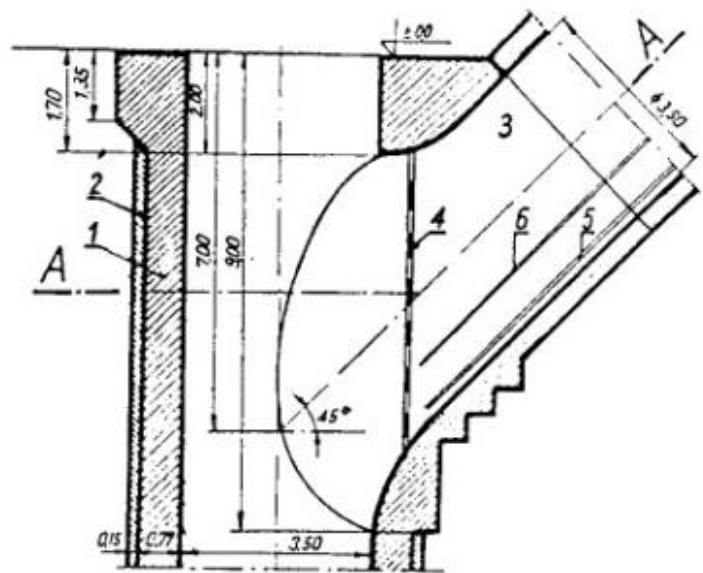


Fig. 17.4.15. Design of ventilation shaft head 12 ft (3.5 m) diameter.

1. hanging brick lining
2. insulation concrete outer ring
3. ventilation channel
4. safety grate
5. ladder
6. railing

Dimensions in cm. Conversion factor: 1 in.
= 2.54 cm.

Shaft Inset I

- *Inset of ventilation shaft with out hoisting gear must be designed to minimize airflow resistance.
- *Size of inset for hoisting shaft depends on width and number of skip, cage being hoisted on this level, number of decks in cage, maximum length of supply to be delivered.
- *All inset section area are to be checked for ventilation requirement.
- *Air velocity at inset should nit exceed permitted level.

Shaft Inset II

- *Inset opening must provide opening as basement (abutment) for pusher and swinging platform, drift for simultaneous entrance and departure by personal from multiple cage, nitch for control equipments, by pass around shaft.
- *Dimension of height of an inset is governed by maximum length of material to be transported.

Shaft Sump I

- *Sump for shaft, equipped for hoisting is more complicated than those of ventilation shaft.
- *Shaft bottom sump is designed to keep following critical item free from water.
 - > Reverse sheave for bottom rope.
 - > End equipment for rigid (stiff) guide.
 - > Weights for rope guides,
- *Shaft bottom sump should remain free from falling materials due to hoisting.
- *Sump should keep shaft hoisting system operable, despite heavy rush of water in shaft.

Shaft Sump II

- *Shaft bottom sump is deepest are and all water getting in shaft system finds way to shaft bottom sump.
- *Shaft bottom sump must provide sufficient capacity, so that operation of shaft does not get disturbed by flooding.
- *Shaft bottom sump must have efficient sump cleaning system to keep hoisting uninterrupted.

Shaft Loading Station I

- *Skip shaft & production levels are connected by a system of opening.
- *Shaft loading arrangement does following function
 - > Flow of material
 - > Sizing material.
 - > Store material
 - > Measure material
 - > Load Material.
- *Shape & size of loading level (chamber) depends on skip size, no of skip, loading system, type of mechanical & electrical services to system.

Shaft Loading Station II

- *Loading inset equipment for hoisting are more complicated than those of other inset.
- *Design of loading level depends on skip loading system, horizontal transport arrangement, loading facilities.
- *Loading level may require
 - > Rail Horizontal transport.
 - > Tippler chamber or unloading ramp chamber (for Granby cars).

Cont

Shaft Loading Station I

Cont

- > Unloading ramp chamber.
 - > Skip chamber > Belt horizontal transport.
 - > Loading Chamber
 - > Retaining bunker
 - > Loading Chamber
 - > Batching chamber.
 - > Skip Chamber
 - > Drift with scale

Shaft Inset I

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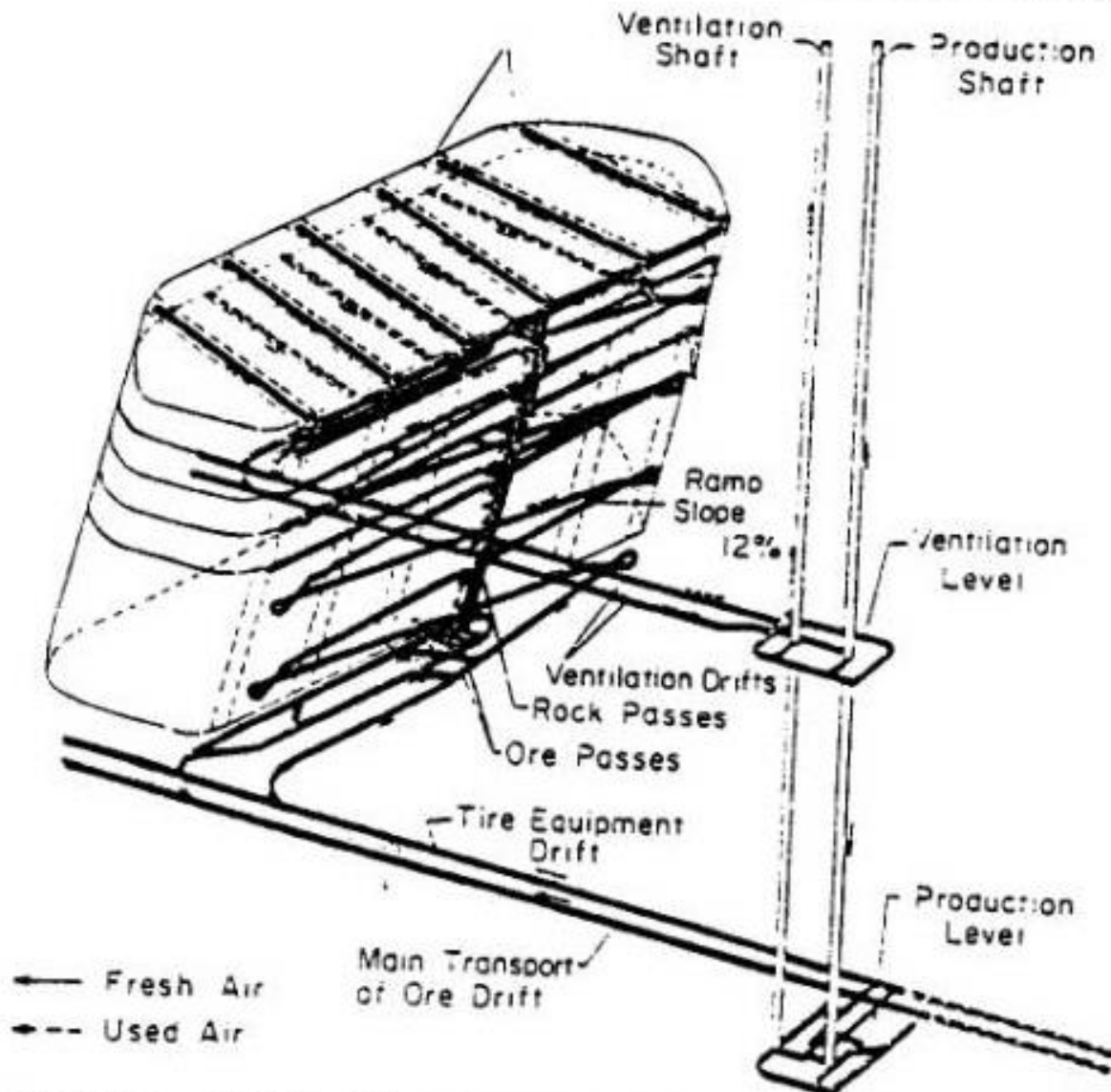


Fig. 17.4.3. Additional development resulting from side-shaft location for a magnetite ore deposit.

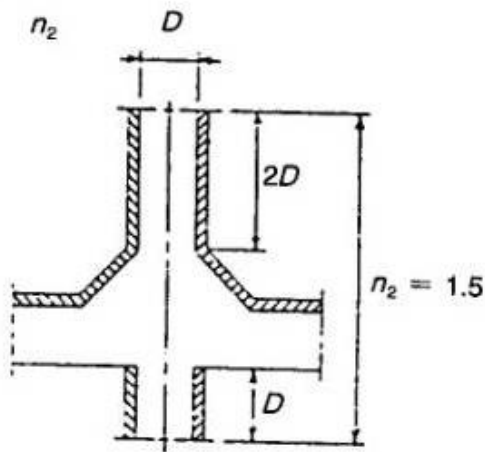
Shaft Inset Stress

Table 17.4.2. Determination of Stress Coefficient n

Symbol	Type	Magnitude
n_0	Influence of tectonic forces	In situ measurements
n_1	Influence of the shaft diameter D	$n_1 = \frac{1}{2} \sqrt[3]{D + 1}$
n_2	Influence of the shaft station	$n_2 = 1.5$
n_3	Influence of the strata dip x	$n_3 = 1.0$ if $x < 30^\circ$ $n_3 = 1.25$ if $x > 30^\circ$
n_4	Influence of the weak strata of thickness h	$n_4 = 1.0$ if $h > 1.3$ m $n_4 = 0.7$ if $0.8 < h < 1.3$ $n_4 = 0.3$ if $h < 0.8$
n_5	Influence of the water filtration through the lining (seepage)	$n_5 = 1.0$ for watertight lining $n_5 = 0.1$ for formation grouting $n_5 = 0.1$ to 0.2 for water drainage $n_5 = 0$ for preliminary lining

Remarks

n_0 —apply coefficient if $\sigma_n / \sigma_v > 1$



n_3 —applies for cohesive soil and rock without water
 n_4 —applies for calculation of $H_{critical}$ in rocks
 n_5 —applies for calculation of water pressure

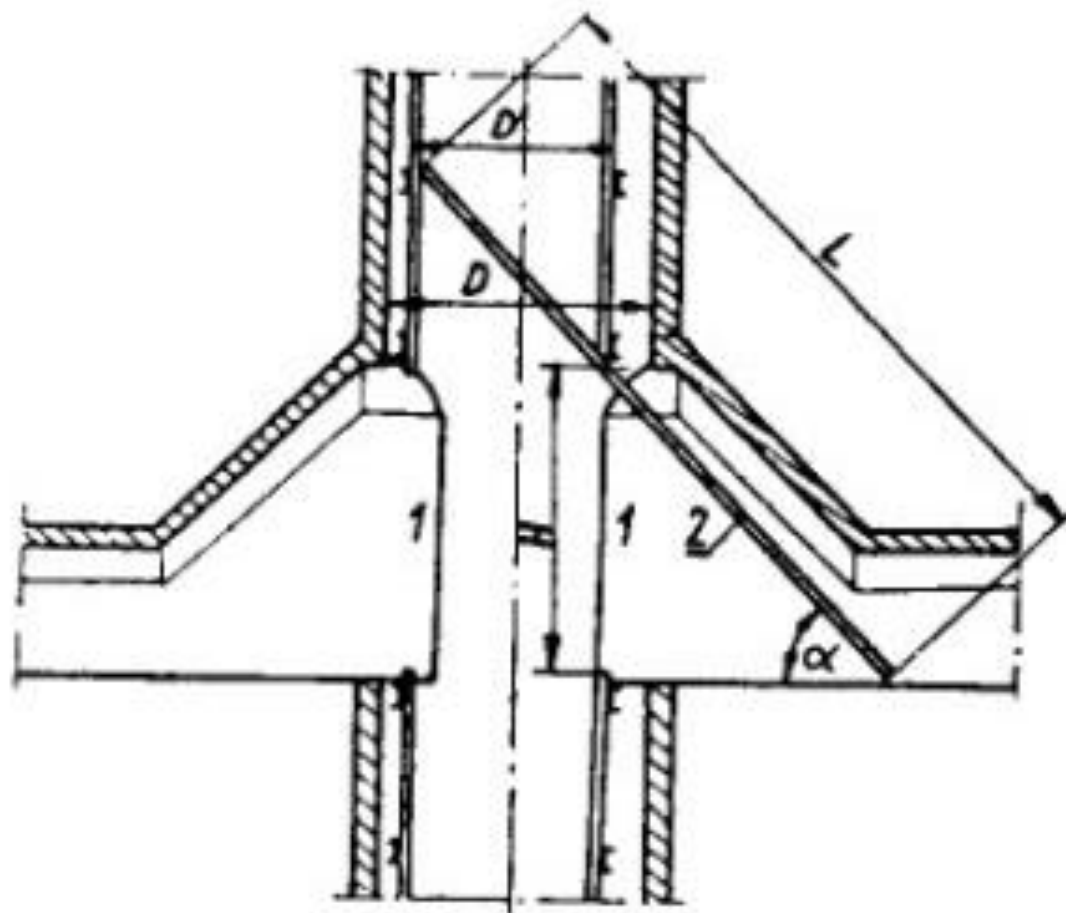


Fig. 17.4.16. Inset height calculation. L is maximum length of transported material, α is angle from level, D is shaft diameter. For $\alpha = 45^\circ$, $D' = 0.7 D$, which is the free space between shaft beams.

$$H = 0.7 (L - D).$$

Shaft Inset I

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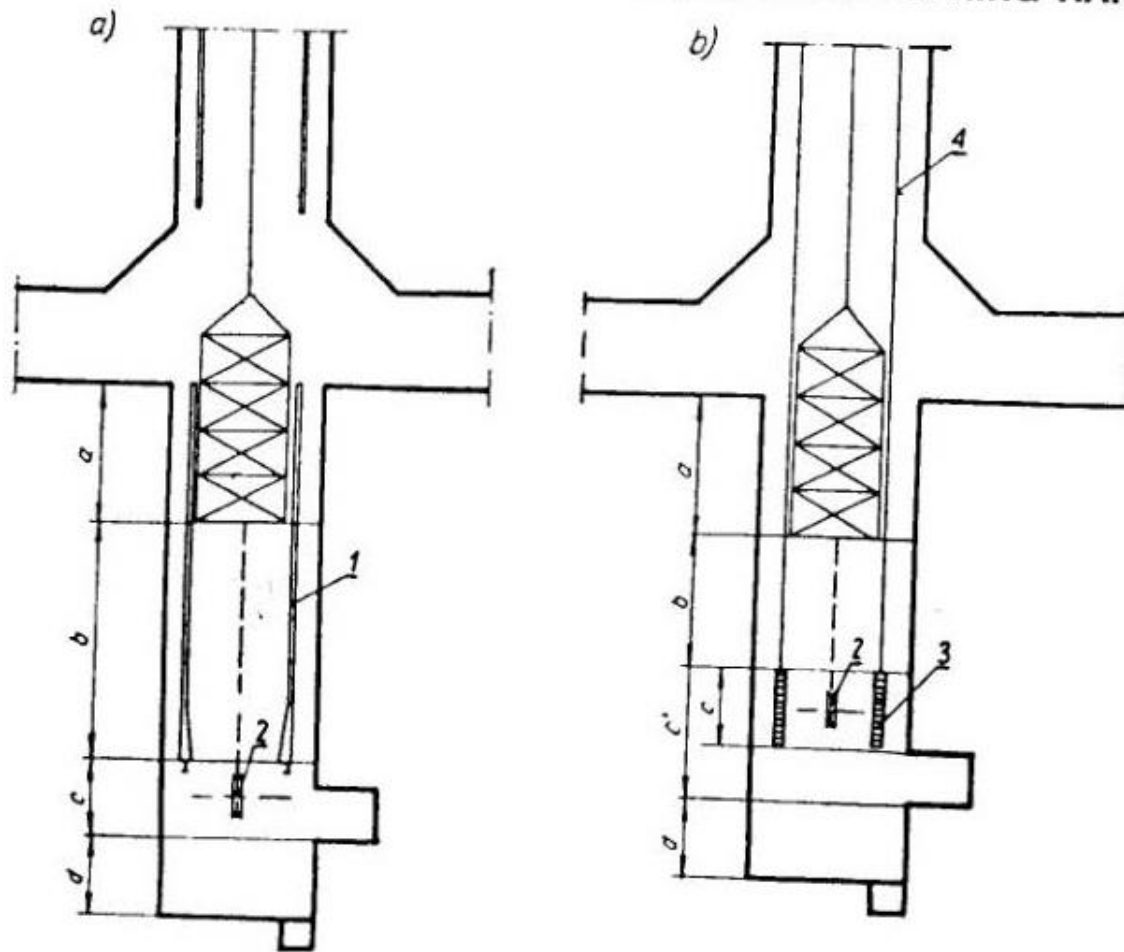


Fig. 17.4.17. Determination of shaft bottom depth (a) in personnel shaft using stiff guides (1), and (b) for material hoisting using rope guides (4). Space must be allowed for return sheave (2) and rope guide weights (3). Then depth (in m) = $a+b+c+d$, where $a = (n - 1)w$, $b = (1-1.5)w$, $c = 2-10$ m, and $d = 3$ m, n is the number of cage floors, w is the distance between floors, v is the maximum hoisting speed in m/s, b is the freeway (max. 35 ft or 10 m), and c is the sheave clearance in m. (After Walewski, 1965.) Conversion factor: 1 ft = 0.3048 m.

Shaft Inset I

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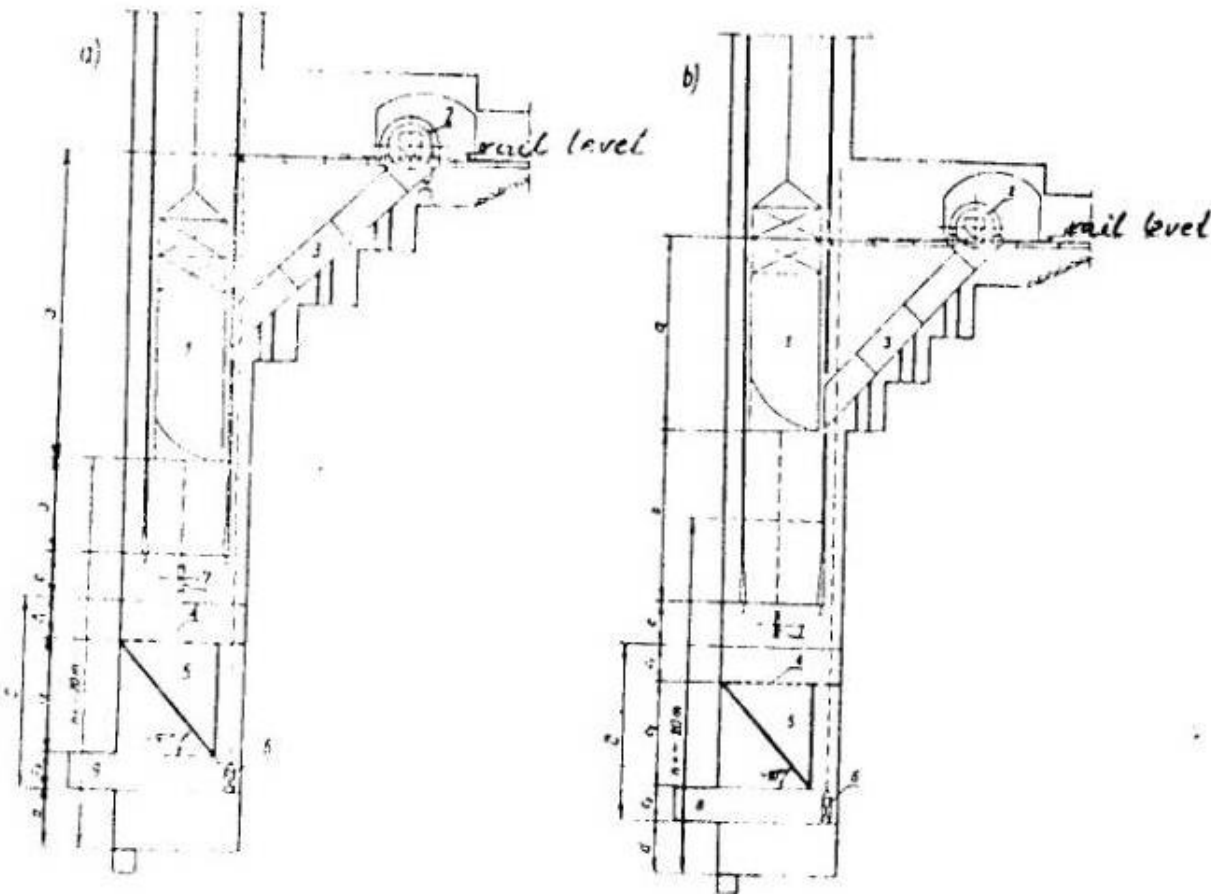


Fig. 17.4.18. Determination of shaft bottom depth in a skip shaft (a) for material hoisting and (b) for personnel hoisting. In this case, depth = $a + b + c + d$, as in Fig. 17.4.17, but a includes additional clearance for the skip (1), tippler (2), and batchers (3); and c includes clearance for the protecting grill (4), spillage bunker (5), auxiliary hoisting bucket (6), and return sheave (7); d is the sump pump chamber (8). (After Walewski, 1965.)

Sinking Preparation I

- *Determining geology & hydrogeology condition of shaft sinking site by drilling a hole or by examining an existing exploratory hole close by.
- *Preparation of technical documentation pertaining to survey & permitting construction work of shaft head.
- *Shaft site should be secured from flooding.
- *100 year old data is collected to define heights flood level of the area.
- *Construction of Roads & material storage area (open & covered).

Sinking Preparation II

- *Grading of terrain for starting sinking operation.
- *Further grading is done at time of sinking by using waste material from shaft.
- *Water supply for drinking, industrial use, fire extinguishing to be established.
- *When water is not available, adequate supply has to be established from other source.
- *Incase of sinking well / bore hole for water supply should be located out side drainage cone of shaft.
- *Water pumped from shaft should be utilized as much as possible.

Sinking Preparation III

- *Approximate requirement of water is provided

- > Industrial water (plant) – 80 to 120m³/day

- > Drinking water (tap) – 40 to 60m³/day

- > Fire tank -- 80 to 100m³

- *Water requirement at each site is to be calculated based on site condition.

- *Adequate storm water drain to be provided at site in planned manner.

- *Garland storm water to be provided around sinking site to prevent flooding.

Sinking Preparation IV

- *Electrical power to be provided in two stages.
- *For preparatory work small transformer 260kva is installed.
- *During sinking 1200kva should be transformer to be commissioned.
- *Transformer should have two independent supply of power.
- *Diesel generator for emergency supply has to be provided.
- *Covered storage for construction material & ramp for unloading heavy machine.

Sinking Preparation V

- *Establishment of telephone communication line.
- *Construction of temporary building for shaft sinking & foundation for head gear.
- *Installation of sinking equipment (sinking winder)
- *Surveying of elevation of shaft collar.
- *Marking main shaft axes & north direction.
- *

Applicability I

1597

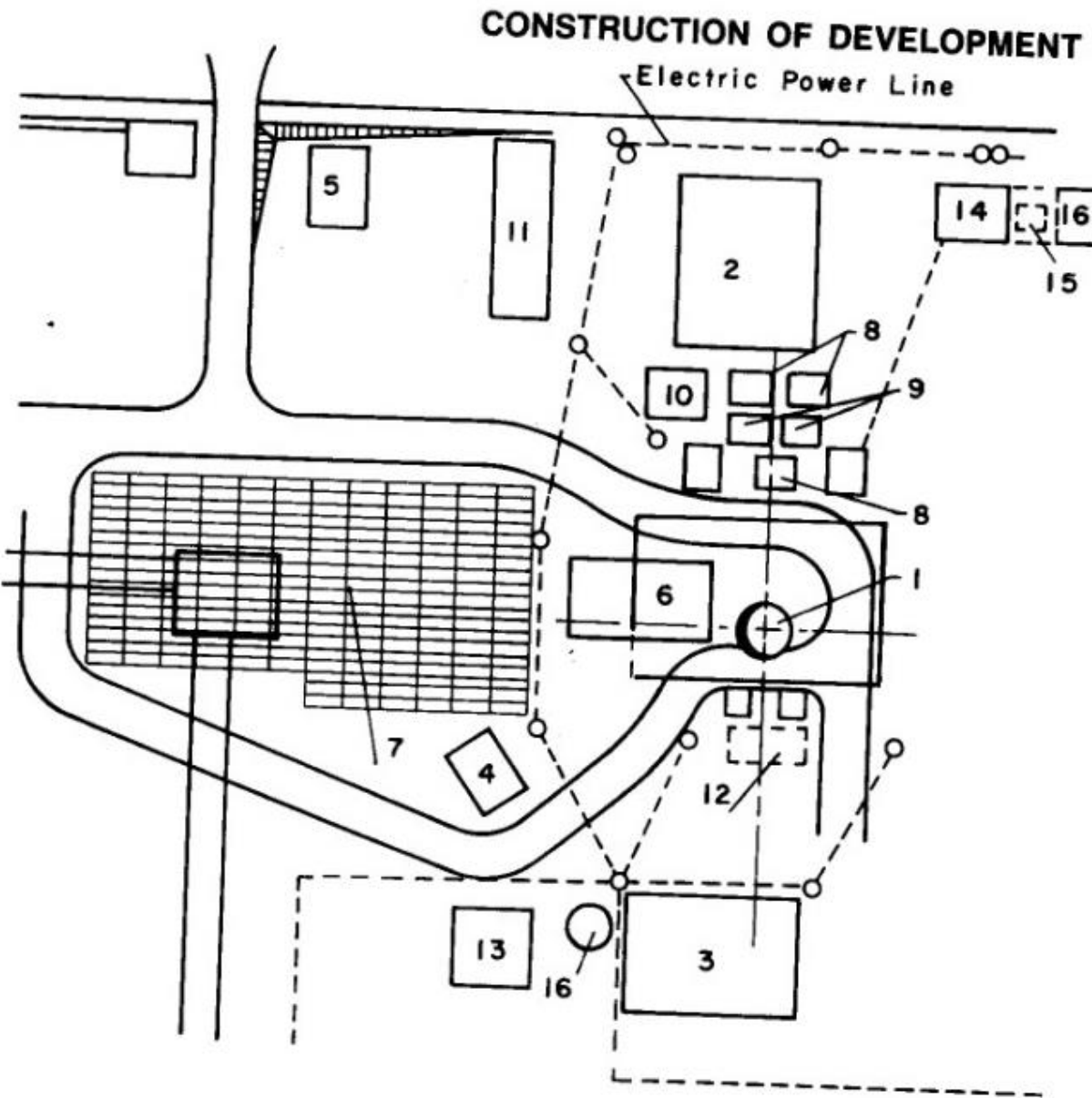


Fig. 17.4.19. Plan of the shaft-sinking site.

1. headframe
 2. hoisting machine house
 3. compressor house
 4. fan
 5. blacksmith shop
 6. concrete plant
 7. open storage for construction materials
 8. reel hoists for concrete form
 9. reel hoist for hanging stage
 10. control house
 11. machine shop
 12. place for emergency reel hoist
 13. water tank for compressor and fire hazard
 14. temporary transformers 6.0/0.5 kV
 15. compressed air tank, 353 ft³ (10 m³) volume
 16. diesel generator
- Conversion factor: 1 ft³ = 0.0283 m³.

Temporary Building I

- *Temporary building should be located so that it shall not interfere with permanent building to be constructed.
- *Building for offices, bath house, first aid room, heating & cooling units, ware house for equipments.
- *Lamp House.
- *Mechanical, Blacksmith shop for repairing work & sharpening bit.
- *Housing of compressor, cooling system,
- *Transformer station, housing transformer, substation switches, electrical workshop.

Temporary Building II

- *Electrical system should be near to maximum load to reduce loss.
- *Housing transformer, substation, switches, electrical shop.
- *Shed to accommodate sinking equipment like drill machine, mucking machines etc.
- *Shed for cement godown & other building material storage space.
- *Room for tugger / stage hoist.
- *Fan house accommodating fan and switch gear etc.

Temporary Building III

- *Oil storage, flammable supply house,
- *Living accommodation for sinking crew & others.

Reference I

*SME Mining Engineering Handbook, 2nd Edition,
Volume II, Page 1580 to 1598.

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